Refraction and Snell’s Law

> SYNOPSIS
>
> A shiny surface sends light back; a transparent one lets it through, but not along the line it arrived on. The bending has one equation, and that equation comes out of something already in hand from Note 1 — that light travels at a different speed in a different medium.

BEFORE YOU START — PREDICT

A ray goes from air into water. Does it bend **toward** the normal or **away** from it — and which quantity in $v = c/n$ decides? Write both down. Figure 1 is drawn to scale, so you can check yourself against it.

PERFORMANCE EXPECTATIONS

1. **State** Snell’s law and identify every symbol in it, including which angle is measured from what.
2. **Predict** which way a ray bends at an interface from the two refractive indices alone.
3. **Calculate** an unknown angle or an unknown index from the other three quantities.
4. **Derive** Snell’s law from the wavelength change, using that the frequency is the same in both media.

4. Law of refraction, Snell’s law

Bending of a light “ray” at an interface between two media of different refractive index.

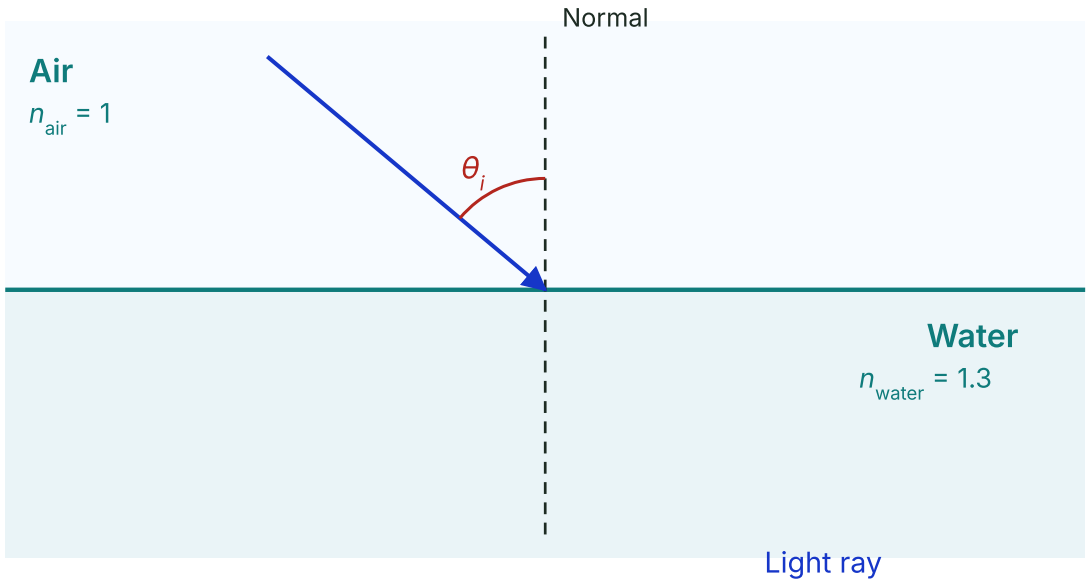


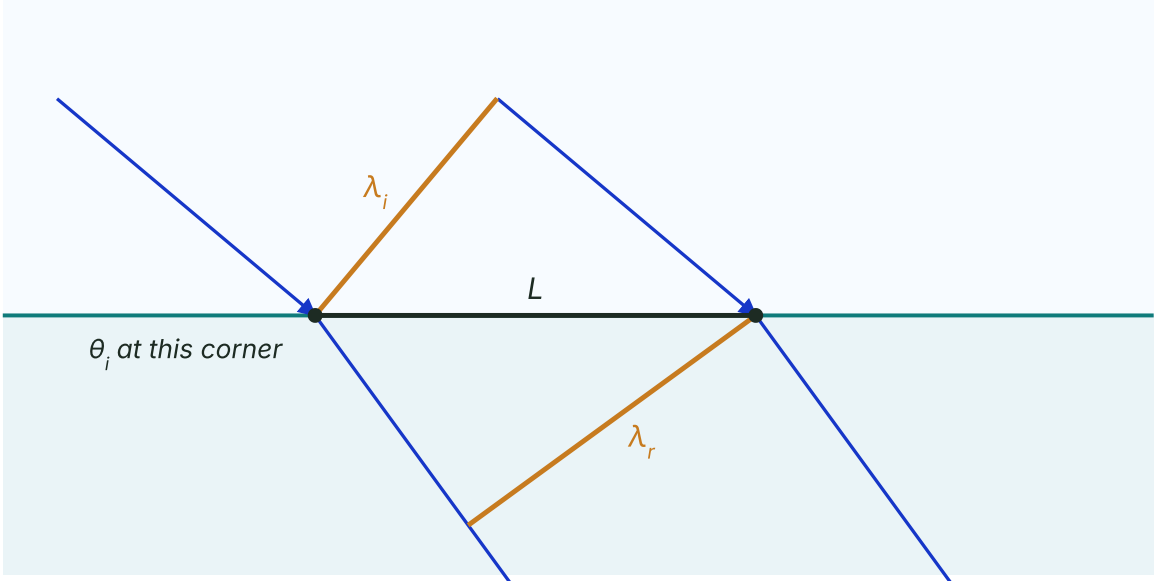
Figure 1. Drawn to scale from the equation itself: $\theta_i = 50°$ in air gives $\theta_r = 36.1°$ in water. Going into the *denser* medium the ray bends **toward** the normal. Both angles are measured from the normal — the same convention as Note 2, and the one the equation is written in.

Your job on Figure 1. The interface, the normal and the incoming ray are drawn for you. Draw the refracted ray on the far side and mark θ_r. Going into the denser medium, does it bend *toward* the normal or away from it?

$$n_i \sin\theta_i = n_r \sin\theta_r$$

Where that equation comes from

Note 1 left us with $v = c/n$: light goes slower in a denser medium. The frequency f does not change when light crosses a boundary — the wave arriving and the wave leaving are the same wave — so if the speed drops, the **wavelength** must drop with it. That is the whole mechanism, and the geometry below turns it into the equation.



Same L, same frequency, two different wavelengths — that is the whole of Snell's law.

Figure 2. The construction behind the equation. Two parallel rays strike the surface a distance *L* apart. λ_i and λ_r are legs of two right triangles that **share the same hypotenuse *L***, so $\sin\theta_i = \lambda_i/L$ and $\sin\theta_r = \lambda_r/L$.

1. Get $n_i \sin\theta_i = n_r \sin\theta_r$ from Figure 2, using $v = f\lambda$ and $v = c/n$.

THE PUNCHLINE

$$n_i \sin\theta_i = n_r \sin\theta_r$$

Bigger *n* means smaller angle from the normal. Nothing in the derivation used anything but the speed change from Note 1 and the fact that frequency survives a boundary — which is why the same equation holds for every pair of transparent media, not just air and water.

Looking ahead

Run the light the other way — out of the dense medium instead of into it — and the angle on the far side grows instead of shrinking. Push it far enough and there is no far side left. **Note 4** is total internal reflection.

2. Which quantity is the **same** on both sides of the boundary, and is the reason the derivation works?

◯ The speed of the light. ◯ The wavelength. ◯ The frequency. ◯ The angle from the normal.

3. A ray passes from water into air. Which way does it bend?

◯ Toward the normal, because it speeds up. ◯ Away from the normal, because it speeds up. ◯ It does not bend; only entering a medium bends light. ◯ Toward the normal, because water has the larger index.

4. Light enters a medium from air at $\theta_i = 45.0°$ and refracts to $\theta_r = 30.0°$. What is the refractive index of that medium?

Check Clear

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